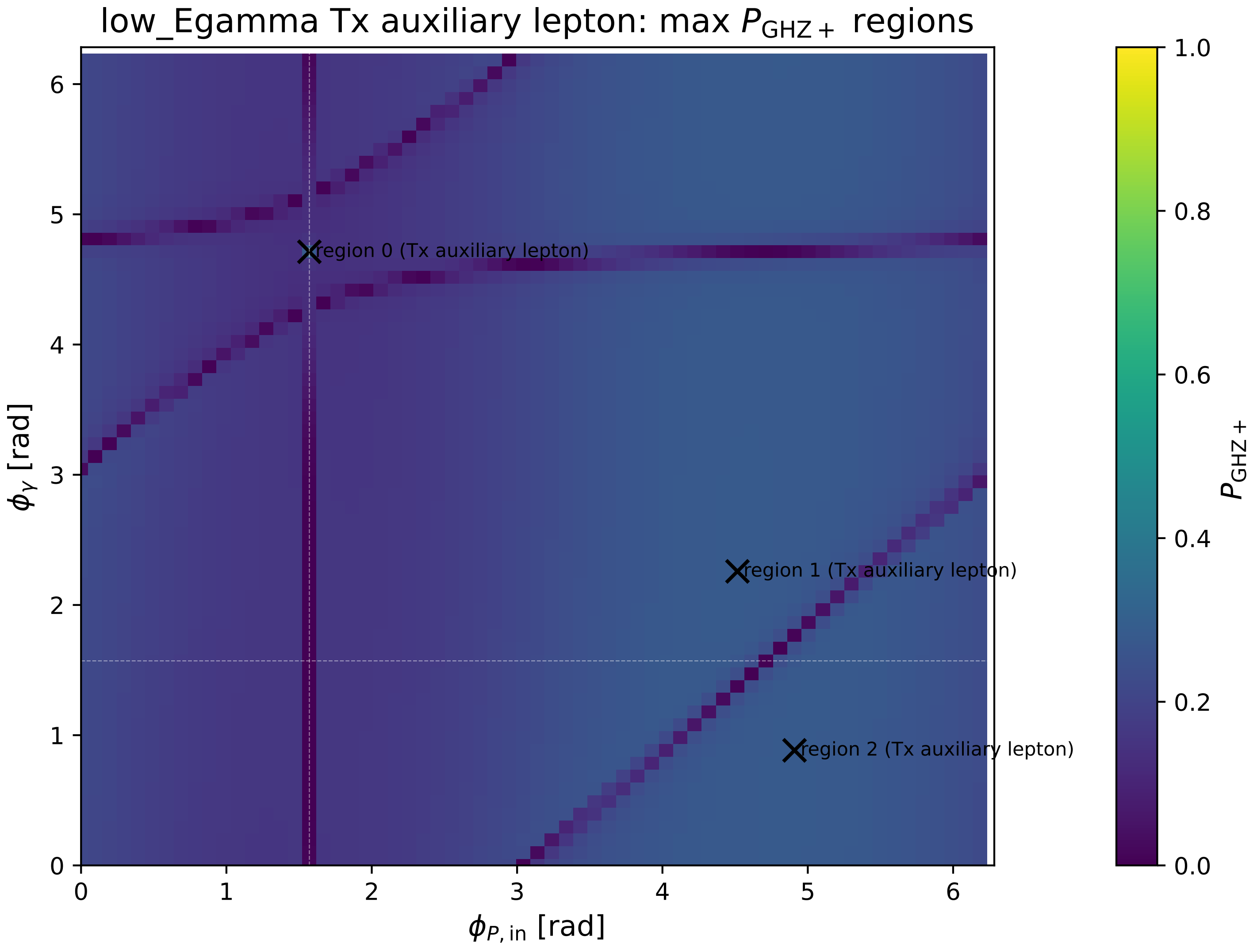
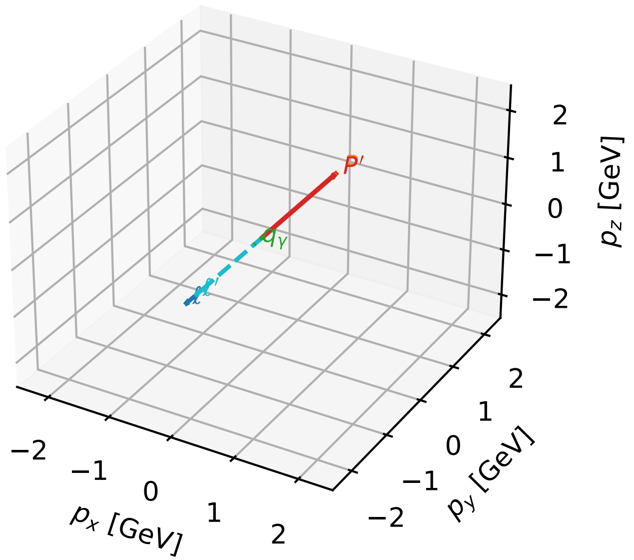


low_Egamma Tx auxiliary lepton: max $P_{\mathrm{GHZ}+}$ regions
region 0 (Tx auxiliary lepton)
region 1 (Tx auxiliary lepton)
region 2 (Tx auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx auxiliary lepton characteristic max $P_{\mathrm{GHZ}+}$ configuration

3D momenta

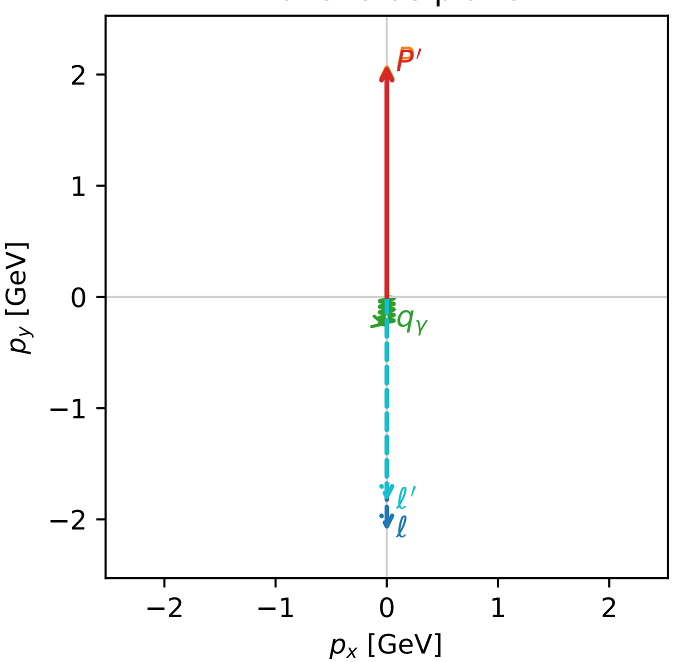


ghz_purity_low_egamma_tx_electron_region_0 P_{\rm GHZ+}=0.279634
region: low_Egamma_Tx_electron_region_0
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\ell}, h_p)$

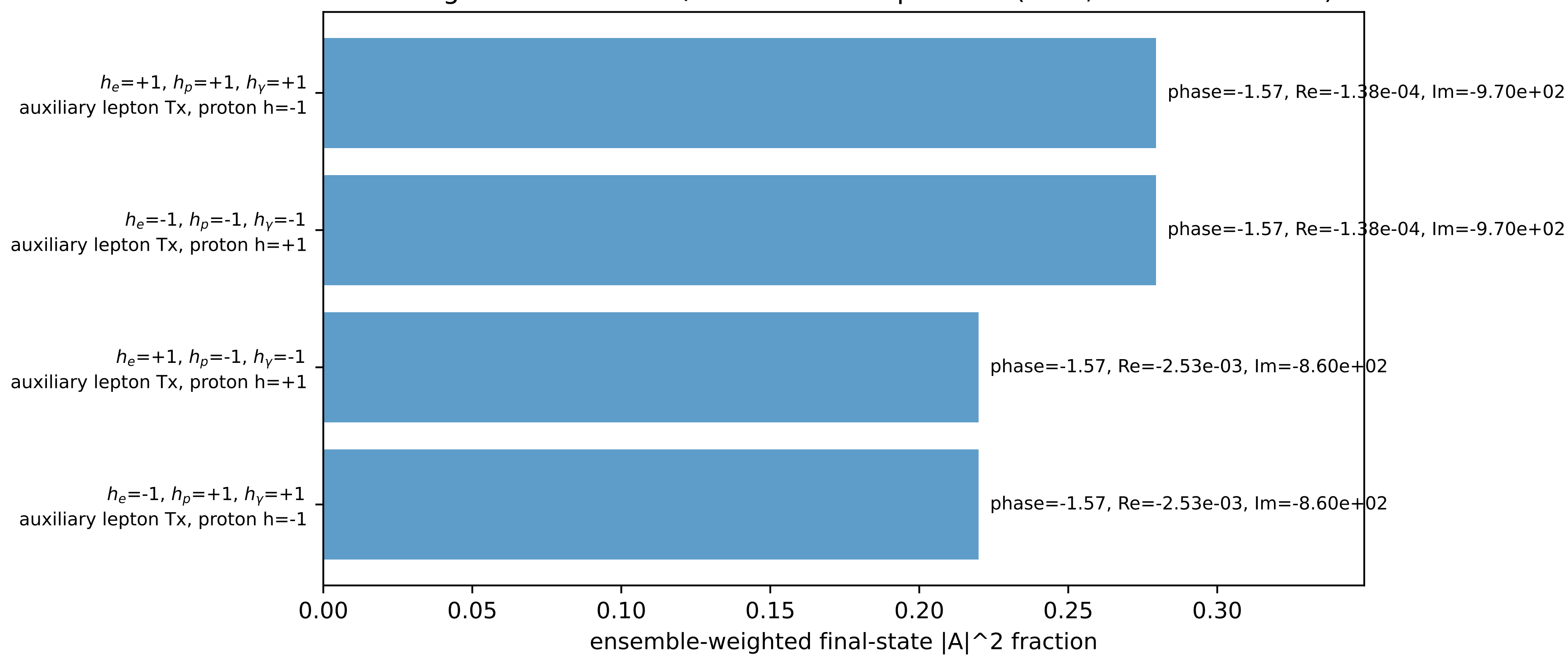
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.09195
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=4.71239
Q^2=0.0143762, x_B=0.00651489, t=-3.73942e-05, W^2=3.07213, y=0.112378
$\theta(\ell', \gamma)$=0 deg

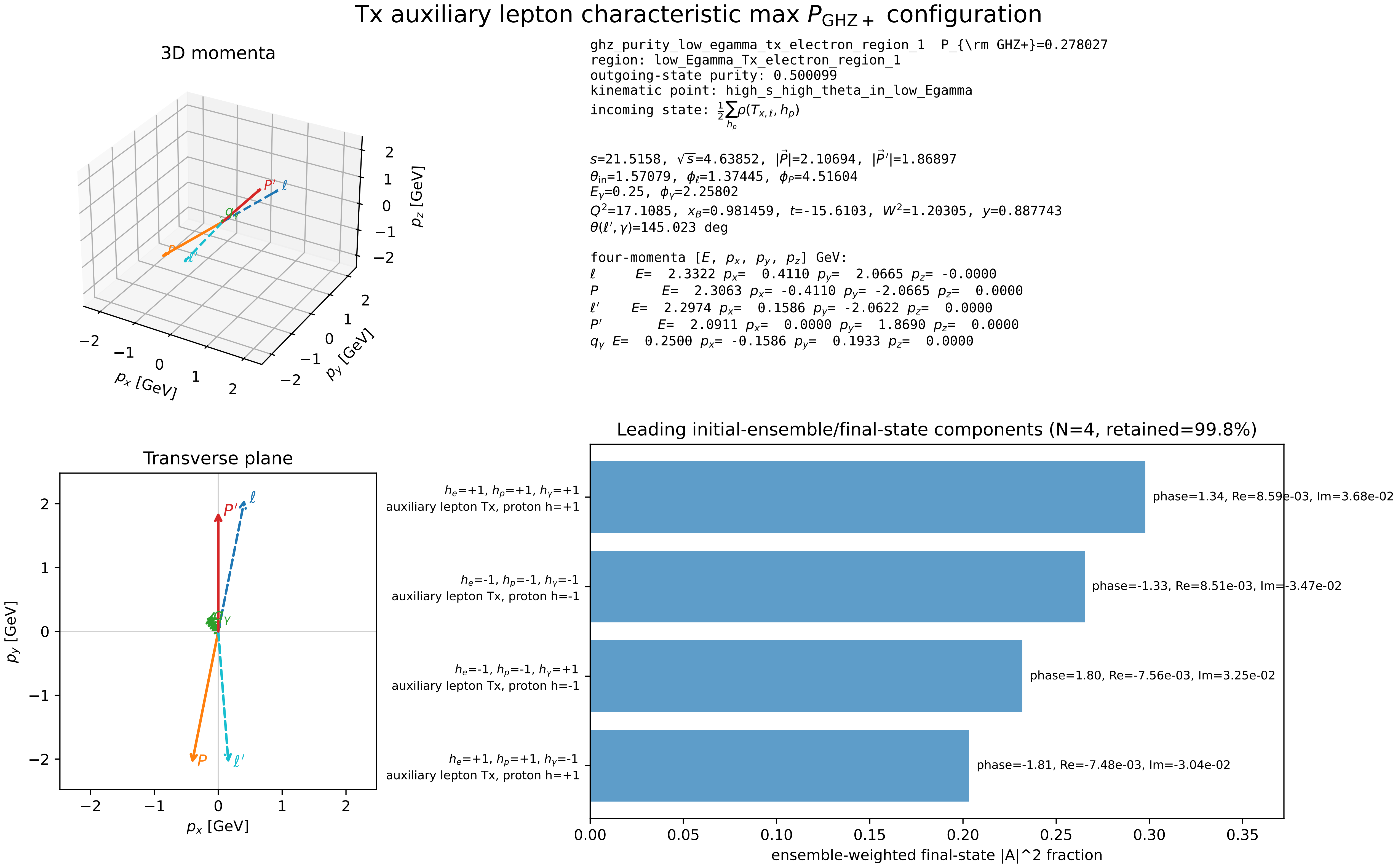
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 2.0959 p_x= 0.0000 p_y= -1.8420 p_z= 0.0000
P' E= 2.2926 p_x= 0.0000 p_y= 2.0920 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0000 p_y= -0.2500 p_z= 0.0000

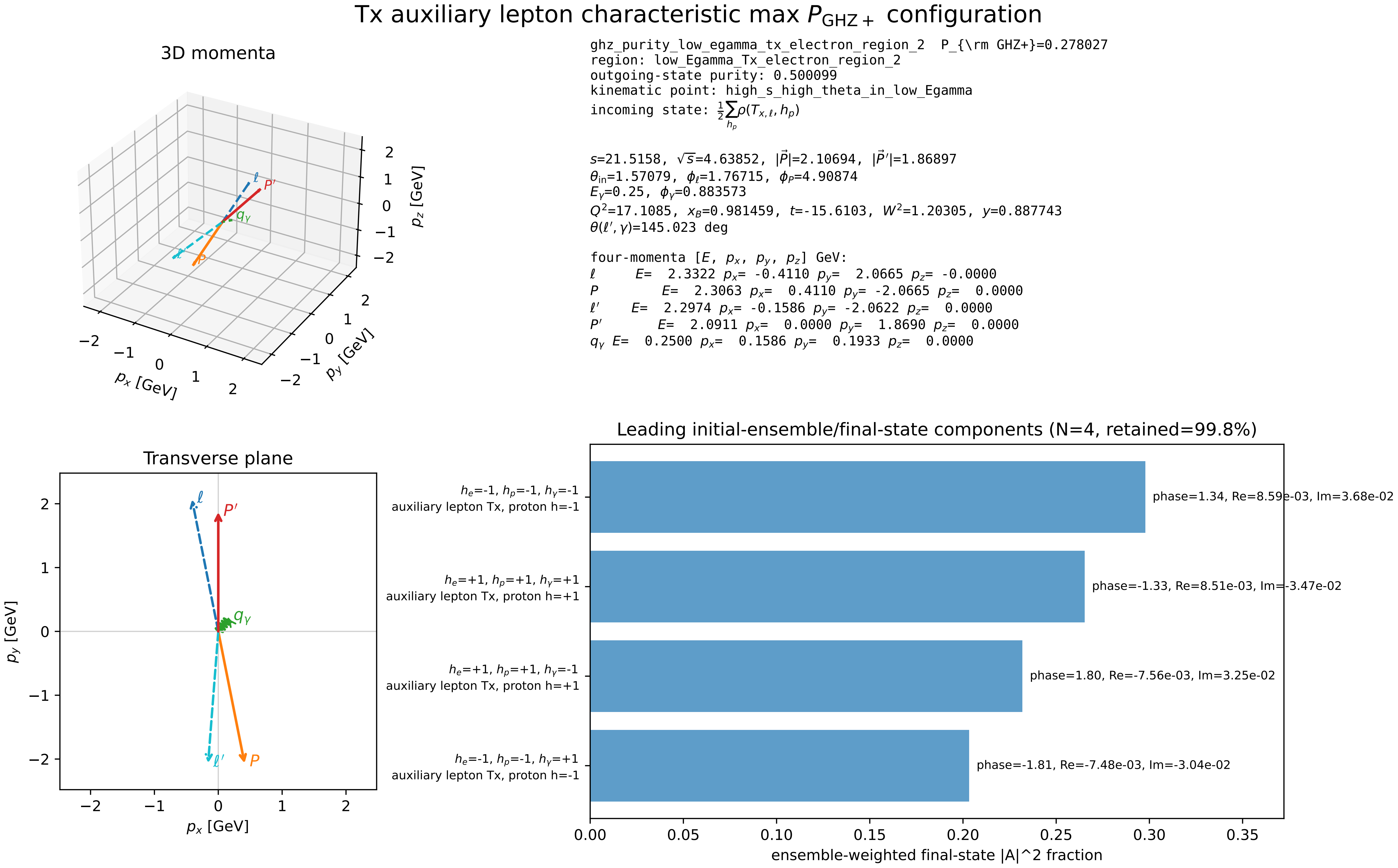
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)




Tx auxiliary lepton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
ℓ
q_γ
P
ℓ'
ghz_purity_low_egamma_tx_electron_region_1 P_{\rm GHZ+}=0.278027
region: low_Egamma_Tx_electron_region_1
outgoing-state purity: 0.500099
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\ell},h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.86897
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.37445, ϕ_P=4.51604
E_γ=0.25, ϕ_γ=2.25802
Q^2=17.1085, x_B=0.981459, t=-15.6103, W^2=1.20305, y=0.887743
$\theta(\ell',\gamma)$=145.023 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.4110 p_y= 2.0665 p_z= -0.0000
P E= 2.3063 p_x= -0.4110 p_y= -2.0665 p_z= 0.0000
ℓ' E= 2.2974 p_x= 0.1586 p_y= -2.0622 p_z= 0.0000
P' E= 2.0911 p_x= 0.0000 p_y= 1.8690 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1586 p_y= 0.1933 p_z= 0.0000
Transverse plane
Leading initial-ensemble/final-state components (N=4, retained=99.8%)
h_e=+1, h_p=+1, h_γ=+1
auxiliary lepton Tx, proton h=+1
phase=1.34, Re=8.59e-03, Im=3.68e-02
h_e=-1, h_p=-1, h_γ=-1
auxiliary lepton Tx, proton h=-1
phase=-1.33, Re=8.51e-03, Im=-3.47e-02
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Tx, proton h=-1
phase=1.80, Re=-7.56e-03, Im=3.25e-02
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Tx, proton h=+1
phase=-1.81, Re=-7.48e-03, Im=-3.04e-02
ensemble-weighted final-state |A|^2 fraction


Tx auxiliary lepton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
P'
q_γ
ℓ'
P
ghz_purity_low_egamma_tx_electron_region_2 P_{\rm GHZ+}=0.278027
region: low_Egamma_Tx_electron_region_2
outgoing-state purity: 0.500099
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\ell}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.86897
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.76715, ϕ_P=4.90874
E_γ=0.25, ϕ_γ=0.883573
Q^2=17.1085, x_B=0.981459, t=-15.6103, W^2=1.20305, y=0.887743
$\theta(\ell', \gamma)$=145.023 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.4110 p_y= 2.0665 p_z= -0.0000
P E= 2.3063 p_x= 0.4110 p_y= -2.0665 p_z= 0.0000
ℓ' E= 2.2974 p_x= -0.1586 p_y= -2.0622 p_z= 0.0000
P' E= 2.0911 p_x= 0.0000 p_y= 1.8690 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1586 p_y= 0.1933 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=99.8%)
h_e=-1, h_p=-1, h_γ=-1
auxiliary lepton Tx, proton h=-1
phase=1.34, Re=8.59e-03, Im=3.68e-02
h_e=+1, h_p=+1, h_γ=+1
auxiliary lepton Tx, proton h=+1
phase=-1.33, Re=8.51e-03, Im=-3.47e-02
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Tx, proton h=+1
phase=1.80, Re=-7.56e-03, Im=3.25e-02
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Tx, proton h=-1
phase=-1.81, Re=-7.48e-03, Im=-3.04e-02
ensemble-weighted final-state |A|^2 fraction